Introduction to Statistical Investigations

Second Edition

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Chapter 5

Comparing Two Proportions
5.3: Theory-Based Approach

Main Idea

- In Exploration Part II, for the murderous nurse, we used the applet to calculate the **significance** and **confidence interval** for relative risk and difference of proportions.
- Here, we will learn equations to compute those ourselves.
- We will focus on difference of proportions.

Standardized Statistic

- Called a two-sample z-test

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}$$

Eurgh. Do I have to memorize that?

- Probably not
  - I would give it to you on an exam
  - But you should know how to recognize it and how to decipher it

Deciphering the standardized statistic: numerator

- $z = \frac{observed\ statistic - hypothesized\ value}{standard\ error\ of\ statistic}$
- What's the hypothesized value for a difference of proportions (#10 on Exploration part I)

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Deciphering the standardized statistic: denominator

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

- Still a square root, that's good, I recognize that.
- What's $\hat{p}$?
- Combined proportion of success
- Add the two conditional proportions together: $\hat{p} = \widehat{p_1} + \widehat{p_2}$

Deciphering the standardized statistic: denominator

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

- We've got $\hat{p}$
- Remember the old standard deviation equation?

$$\sqrt{\frac{\pi(1-\pi)}{n}}$$

- Remember we replaced π with $\hat{p}$ for the confidence interval, since we don't know all that information for the population?

$$\hat{p} \pm \text{multiplier} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Deciphering the standardized statistic: denominator

- We still have $\hat{p}(1- \hat{p})$ in the numerator of the SD square root.
- We still divide by the sample size
- We just have to add the two sample sizes together in the denominator, since the sample sizes are different.

$$\boxed{z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\left(1 - \hat{p}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}}$$

What about the Confidence Interval?

- The margin of error will be *slightly* different from the SD.
- We are trying to estimate the size of the difference between population proportions.
- We are under the assumption there is a difference!

- Rather than in the standardized statistic, where we are assuming the null hypothesis is true.

Ok now give me the equation

- $Margin\ of\ error\ =\ multiplier \times \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

- The standard error (square root bit) still has the observed * 1-observed in the numerator
- Still dividing by the sample size in the denominator
- We're just not combining $\hat{p}_1$ and $\hat{p}_2$ this time

- This will also be given to you on an exam

Let's do an example: Murderous Nurse

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$$

- $\hat{p}_1 =$
- $n_1 =$

- $\hat{p}_2 =$
- $n_2 =$

- $\hat{p} = \hat{p}_1 - \hat{p}_2 =$
- $z =$

Let's do an example: Murderous Nurse

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$$

- $\hat{p}_1 = 0.156$
- $n_1 = 257$

- $\hat{p}_2 = 0.025$
- $n_2 = 1384$

- $\hat{p} = \hat{p}_1 - \hat{p}_2 = 0.131$
- z = (0.156 – 0.025) / sqrt(0.131(0.869)(0.00389+0.0007225)) = 0.131/0.0229 = 5.72

Let's do an example: Murderous Nurse

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

- $\hat{p}_1 =$ $\hat{p}_2 =$
- $n_1 =$ $n_2 =$

$$\text{Margin of error} = \text{multiplier} \times \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

- $\hat{p} = \hat{p}_1 - \hat{p}_2 =$
- SE =
- Multiplier = 2
- $\hat{p} \pm 2\ SE =$

Let's do an example: Murderous Nurse

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

- $\hat{p}_1 = 0.156$ $\hat{p}_2 = .025$
- $n_1 = 257$ $n_2 = 1384$

$$Margin\ of\ error = multiplier \times \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

- $\hat{p} = \hat{p}_1 - \hat{p}_2 = 0.131$
- SE =
- Multiplier = 2
- $\hat{p} \pm 2\ SE$ =

Validity Conditions: When can you use these equations?

- At least 10 observations in each cell
- Still abides by our 10 successes and 10 failures rule